

School of Computer Science and Engineering

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Project Details						
Project Title:		AyurDietOS – Smart Ayurvedic Diet Management				
Type of Project		Web-Based Software Project / Cloud-Based Application				
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Contents

- Abstract
- 1. Introduction
- 2. Literature Survey
- 3. Positioning
 - 3.1 Problem statement
 - 3.2 Product position statement
- 4. Stakeholder Descriptions
 - 4.1 User stakeholders
- 5. Project overview
 - 5.1 Objectives
 - 5.2 Goals
 - 5.3 Feasibility Study
 - 5.4 Alternatives
 - 5.5 Budget
 - 5.6 Key deliverables
 - 5.7 Necessary materials
 - 5.8 Methodology
 - 5.9 Modules identified
- 6. Requirements and cost Analysis
- 7. Developing project execution timelines
- 8. Survey Paper Publication details
- 9. Conclusions
- 10. References

Abstract:

AyurDietOS is a cloud-based Ayurvedic diet intelligence and patient management system designed to assist dietitians in creating personalized and holistic diet plans. In traditional Ayurvedic practice, diet charts are often handwritten and manually maintained, which can be time-consuming, inconsistent, and prone to errors. Modern nutrition software, while effective in tracking nutrients, fails to incorporate Ayurvedic principles such as Rasa (taste), Guna (qualities), Virya (potency), and Dosha effects, creating a gap in practice. This project addresses that gap by combining modern nutrient analysis with Ayurvedic dietary concepts to provide a comprehensive platform for diet planning. AyurDietOS allows practitioners to maintain detailed patient profiles, including health parameters, dietary habits, and lifestyle data, while also providing a large database of foods and recipes categorized by both nutrient content and Ayurvedic properties. The system automates the generation of diet charts tailored to individual needs and supports printable reports for patient guidance. Through this project, we aim to enhance the accuracy, efficiency, and practicality of Ayurvedic diet management, making it easier for practitioners to deliver scientifically sound, personalized, and holistic dietary care. The study concludes that integrating modern nutrient science with Ayurveda in a cloud-based platform can significantly improve patient care and streamline dietary practice for professionals.

1. Introduction:

Ayurveda, the traditional system of medicine originating in India, emphasizes the importance of a balanced diet and proper nutrition in maintaining health and preventing disease. According to Ayurvedic principles, every individual has a unique constitution (Prakriti) and may have imbalances in one or more of the three doshas—Vata, Pitta, and Kapha. These dosha imbalances, along with lifestyle, seasonal changes, and health conditions, strongly influence dietary requirements. Therefore, creating personalized diet plans tailored to each individual is a cornerstone of Ayurvedic practice. Traditionally, diet charts in Ayurvedic hospitals and clinics are prepared manually by doctors and dietitians, often written by hand or maintained in spreadsheets. This manual approach is time-consuming, prone to errors, difficult to update, and lacks consistency in tracking patient progress over time.

In parallel, modern nutrition software solutions exist that focus on calorie counting, macronutrients, and micronutrients, but they rarely account for the holistic Ayurvedic perspective. As a result, there is a gap between modern dietary analysis and the traditional principles of Ayurveda. This gap creates challenges for dietitians who wish to provide accurate, patient-specific dietary guidance that respects both modern nutrition science and Ayurvedic philosophy.

The motivation for developing AyurDietOS comes from the need to bridge this gap by providing a cloud-based, user-friendly platform that integrates modern nutrient analysis with Ayurvedic dietary concepts such as Rasa (taste), Guna (qualities), Virya (potency), and their effects on Doshas. The platform will allow dietitians to maintain detailed patient profiles, manage a comprehensive food and recipe database, and generate personalized diet charts efficiently. By automating repetitive tasks, improving accuracy, and providing a structured system for Ayurvedic diet management, AyurDietOS aims to enhance patient care and simplify the workflow of Ayurvedic practitioners.

Through this project, students will demonstrate how technology can be leveraged to support traditional healthcare practices while improving efficiency, accuracy, and scalability. The system is envisioned as a practical tool for Ayurvedic hospitals, clinics, and individual dietitians, enabling them to provide holistic, scientifically-informed dietary guidance that aligns with the principles of Ayurveda.

2. Literature Survey:

In recent years, there has been increasing interest in integrating traditional medical knowledge systems such as Ayurveda with modern digital nutrition and health-management technologies. Early research in this domain primarily focused on digitizing Ayurvedic dietary principles and exploring ways to personalize diet charts based on a patient's prakriti (body constitution). Sharma et al. (2015) [1] examined some of the earliest digital tools for Ayurvedic diet planning, highlighting the limitations of manual chart preparation, which was often inconsistent and difficult to scale for larger patient populations.

Kumar and Singh (2017) [2] later developed a functional prototype of an Ayurvedic diet management system capable of automating basic dosha-based recommendations and maintaining patient records. While this system addressed issues of manual error and data inconsistency, it lacked deeper nutrient analysis and did not incorporate a comprehensive food database.

Advancements continued with Reddy et al. (2019) [3], who proposed a hybrid nutrition platform that integrated macro- and micro-nutrient analysis with traditional Ayurvedic food classifications such as Rasa, Guna, and Virya. Their results demonstrated improved adherence to Ayurvedic recommendations and laid the foundation for more robust digital Ayurveda-based nutrition systems.

Despite these contributions, existing solutions still fall short of offering a cloud-based, fully integrated platform combining patient management, large-scale food databases, and automated diet chart generation rooted firmly in Ayurvedic principles. This gap motivates the development of AyurDietOS, a scalable, user-friendly cloud platform that bridges modern nutrient science with classical Ayurvedic dietary guidelines.

Consolidated Literature Survey of Existing Ayurvedic and AI-Based Diet Systems

Paper Title	Idea / Objective	Dataset Used	Algorithms / Techniques	Performance / Accuracy	Result / Outcome	Drawbacks / Limitations
Ayurvedic Diet vs Standard Diet (2013)	Obesity dietary intervention trial	40 patients	Ayurvedic diet + statistics	Significant TG & VLDL reduction	More effective diet	Short 8-day duration
Diet Recommendation Using Fuzzy Ontology & Type-2 Fuzzy Logic (2016)	Fuzzy logic + ontology diet	Ayurvedic rules	Type-2 fuzzy + ontology	≈94% (claimed)	Season-dosha diet	No validation
Ayurvedic Knowledge Review (2016)	Foundations of Ayurvedic dietetics	Classical texts	Conceptual	None	Strong theoretical basis	No technical implementation
IoMT Diet Recommendation (2020)	ML-based clinical diet	30 patients + 1000 foods	LR, NB, MLP, RNN, LSTM	LSTM=97.74 %	Predicts allowed foods	Small dataset
Digital Ayurveda EHR (2022)	Standardized Ayurvedic digital records	Practitioner interviews	Conceptual architecture	None	EHR framework	No implementation
Food Recommender Prakriti (2022)	DL-based prakriti-season diet	Interview recall data	CNN/RNN/LSTM	None	Concept design	No evaluation
AI Ayurvedic Nutritional App (2022)	Scan food → Ayurvedic value	16 interviews + knowledge base	AI + CV + gamification	None	Concept app idea	Small sample size
Heart Disease Diagnosis & Diet Recommendation Using Ayurvedic Dosha Analysis (2023)	AI + pulse-based dosha analysis	Internal pulse dataset	LR, RF, DT, KNN	DT=99.7%	Accurate dosha + diet	Overfitting, no dataset
Hybrid Diet Recommender (2023)	Nutrient clustering diet	Nutrient database	K-means + RF	None	BMI-based varied diet	No validation
AyuAahar SaaS Plan (Year Not Specified, Placed Here)	Cloud Ayurvedic diet SaaS	Ayurvedic Knowledge Graph	AI + Microservices	None	Software blueprint	No validation
Smart Diet India (2025)	AI + Indian cuisine + Ayurveda	10k recipes + 50k images	ResNet50, NLP, CF	92% detection accuracy	High adherence	Limited

3. Positioning

3.1. Problem statement: In modern Ayurvedic practice, diet charts are often created manually, which is time-consuming, inconsistent, and prone to errors. Existing nutrition software focuses on macro- and micro-nutrient tracking but does not account for Ayurvedic dietary principles such as Rasa, Guna, Virya, and Dosha effects. This gap makes it difficult for dietitians to deliver personalized, holistic, and accurate dietary care efficiently.

3.2. Product position statement: AyurDietOS is a cloud-based Ayurvedic diet intelligence and patient management system designed for Ayurvedic dietitians, clinics, and hospitals. It integrates modern nutrient analysis with Ayurvedic dietary principles to create accurate, personalized, and balanced diet charts. By simplifying patient management, automating diet planning, and providing a comprehensive food database, AyurDietOS helps practitioners deliver effective, holistic nutritional guidance to their patients with ease and efficiency.

4. Stakeholder Descriptions

Project stakeholders are individuals or organizations who are either actively involved in the development of AyurDietOS or whose interests may be affected by its implementation. These stakeholders have a direct or indirect interest in the project outcome and may benefit from its successful completion or be impacted if the project does not meet its goals.

4.1. User stakeholders: The primary user stakeholders of AyurDietOS are the practitioners and beneficiaries of Ayurvedic diet management:

1. Ayurvedic Dietitians:

- Will use the platform to create, manage, and customize patient-specific diet charts.
- Benefit from automated nutrient analysis combined with Ayurvedic dietary principles, reducing manual workload and improving accuracy.

2. Ayurvedic Doctors & Clinicians:

- Can monitor patient dietary plans and ensure that prescribed diets align with overall treatment goals.
- Benefit from easy access to patient profiles and historical diet charts for better decision-making.

3. Patients:

- Receive personalized diet charts and guidance based on their health parameters, dosha profiles, and nutritional needs.
- Benefit from clearer instructions and a structured, holistic dietary plan to support health outcomes.

4. Hospitals & Ayurvedic Clinics:

- Benefit from streamlined diet management workflows and improved patient care efficiency.
- Can maintain digital records, generate reports, and integrate the system with existing hospital management systems.

5. Project overview:

AyurDietOS is a cloud-based platform designed to assist Ayurvedic dietitians in creating personalized, accurate, and holistic diet plans for patients. In traditional Ayurvedic practice, diet charts are often handwritten or maintained using basic spreadsheets, which are time-consuming, prone to errors, and difficult to update or track over time. While conventional nutrition software exists, it primarily focuses on macro- and micro-nutrient analysis and does not consider Ayurvedic dietary principles such as Rasa (taste), Guna (qualities), Virya (potency), and Dosha effects. AyurDietOS bridges this gap by providing a unified, cloud-based system that integrates modern nutritional analysis with Ayurvedic concepts, allowing dietitians to manage patient information efficiently, generate personalized diet charts, and deliver holistic dietary guidance. The platform is aimed at improving patient outcomes, streamlining practitioner workflows, and ensuring that traditional Ayurvedic principles are accurately applied in diet planning.

5.1. Objectives: The primary objective of AyurDietOS is to develop a fully functional cloud-based system that enables Ayurvedic dietitians to create patient-specific diet plans efficiently and accurately. The system will integrate modern nutrient analysis with Ayurvedic principles to ensure balanced, dosha-aligned diet recommendations. It will automate the creation of diet charts, maintain detailed patient profiles, provide access to a large food and recipe database, and generate printable reports for patients. This approach reduces manual effort, enhances the accuracy of dietary guidance, and ensures consistency across different practitioners.

5.2. Goals: The goal of the project is to provide a user-friendly, cloud-based platform that supports dietitians in delivering scientifically informed and holistic nutrition guidance. By simplifying patient management, automating diet chart generation, and maintaining a comprehensive food database with Ayurvedic properties, the system aims to enhance practitioner efficiency, improve patient satisfaction, and bridge the gap between traditional dietary planning and modern nutritional science.

5.3. Feasibility Study: AyurDietOS is highly feasible as a student project. Economically, no significant costs are involved since the project will be developed using open-source software, free web development tools, and personal computing resources. Technically, the project leverages modern web technologies and cloud infrastructure, which are sufficient to meet all functional requirements. Legally, patient data will be conceptualized as stored securely, complying with standard privacy practices. Scheduling feasibility has been considered, and the project is achievable within the academic semester timeline allocated for final-year major projects.

5.4. Alternatives: Currently, Ayurvedic diet charts are either handwritten or maintained using basic spreadsheets, which are time-consuming and prone to human errors. Conventional nutrition software exists, but it primarily focuses on macronutrients and micronutrients without considering Ayurvedic principles such as dosha balance, Rasa, Guna, and Virya. AyurDietOS addresses these limitations by providing an integrated, cloud-based solution that combines both modern nutrition and traditional Ayurvedic dietary concepts.

- 5.5. Budget:** Commercial deployment would require significant capital ranging from ₹25 Lakhs (lean startup) to ₹3 Crores (full-scale deployment) in the first year. The recommended approach is a phased deployment starting with ₹10-15 Lakhs for MVP validation, followed by staged investment based on market traction. This strategy balances the potential for significant impact in Ayurvedic healthcare management with prudent financial risk management.
- 5.6. Key deliverables:** The key deliverables of AyurDietOS include a cloud-based application, a patient and food database with Ayurvedic and nutritional properties, an automated diet chart generation module, printable diet charts for patients, and complete project documentation summarizing system design and methodology.
- 5.7. Necessary materials:** The project requires a computer or laptop with internet connectivity, modern web development software and IDEs, a cloud-based database management system for data storage and authentication, frontend hosting platform, open-source development frameworks and libraries, a comprehensive food database with nutritional and Ayurvedic properties, and digital tools for documentation purposes.
- 5.8. Methodology:** The project will follow a systematic methodology beginning with requirement gathering and analysis, followed by system and database design. The development will be modular, with separate modules for Patient Management, Food & Recipe Database, Diet Chart Generation, Reporting, and User Interface. Each module will be developed and tested individually, then integrated into a complete system. After integration, the platform will undergo functional testing and validation before deployment on a cloud platform. Finally, documentation and project reporting will summarize the development process, system features, and outcomes.
- 5.9. Modules identified:** The project is divided into several high-level modules. The Patient Management Module manages patient profiles and health information. The Food & Recipe Database Module stores and categorizes food items along with their nutritional and Ayurvedic properties. The Diet Chart Generation Module automatically creates personalized diet plans based on patient data and Ayurvedic principles. The Reporting Module generates printable diet charts and reports for patients. The User Interface Module provides an intuitive web or mobile interface for dietitians and patients to interact with the system efficiently.

6. Requirement and cost Analysis

For commercial deployment of the Ayurvedic Diet Management System in hospitals and clinics, the solution requires production-grade cloud infrastructure, secure database services, and a small but dedicated technical and support team. The system must ensure high availability, data security, and regulatory compliance, while remaining scalable as more institutions are onboarded. The major cost components include personnel (development, DevOps, and support), cloud hosting and database services, legal and compliance activities, marketing and sales for client acquisition, and operational overheads such as office space, hardware, and communication. The following table summarizes an approximate annual cost estimate for a limited rollout to about 5–10 hospitals or clinics.

Sl. No.	Cost Component	Description	Estimated Annual Cost (₹)
1	Personnel	1 full-stack developer, 1 DevOps/IT engineer, 1 support executive, part-time Ayurvedic/clinical expert	14,00,000
2	Cloud Infrastructure	Application hosting, database, backups, monitoring, domain and SSL certificates	2,50,000
3	Legal & Compliance	Company compliance, basic security audit, documentation, insurance	2,70,000
4	Marketing & Sales	Website, digital marketing, hospital visits, demos, basic promotional material	3,00,000
5	Training & Onboarding	Staff training sessions, user manuals, initial implementation support	1,00,000
6	Operations & Overheads	Co-working/office space, hardware upgrades, internet, utilities, miscellaneous expenses	2,50,000
	Total Estimated Annual Cost		25,70,000 (≈ ₹25–26 Lakhs)

7. Developing project execution timelines.

Phase 1: Problem Exploration & Initial Ideation (09 Sept – 24 Sept 2025)

09 September 2025

- Shortlisted 5–6 potential problem statements from SIH and other innovation platforms.
- Conducted internal discussions to evaluate feasibility and innovation scope.

14 September 2025

- Reviewed relevant patents to ensure novelty and uniqueness.
- Each team member analyzed assigned documents and shared insights.

17 September 2025

- Continued refining the ideas and explored additional innovative directions.
- Discussed alignment with team strengths in AI, Data Science, and Web Development.

24 September 2025

- Finalized one project problem statement after comparing feasibility, relevance, and academic value.
- Clearly divided early conceptualization roles within the team.

Phase 2: Requirement Analysis & Concept Finalization (28 Sept – 15 Oct 2025)

28 September 2025

- Developed the final problem statement: “Comprehensive Cloud-Based Ayurvedic Practice Management & Nutrient Analysis System.”
- Outlined system modules, features, and user roles (admin, doctor, patient).

15 October 2025

- Planned the overall system architecture (frontend and backend).
- Assigned responsibilities for UI/UX, backend development, database design, and Ayurvedic-rule integration.
- Prepared initial workflow diagrams for the proposed system.

Phase 3: Guided System Refinement & Domain Validation (23 Oct – 25 Oct 2025)

23 October 2025

- Updated system design and functional workflows based on faculty feedback.
- Refined module interactions to better reflect real-world doctor–patient processes.

25 October 2025

- Received guidance to consult an Ayurvedic practitioner for domain validation.
- Initiated research on Ayurvedic diet principles such as dosha, virya, rasa, and guna.
- Mapped how these traditional elements would translate into digital logic.

Phase 4: Prototype Development & Research Consolidation (22 Nov 2025)

22 November 2025

- Completed UI/UX prototype for:
 - Admin Dashboard
 - Doctor Interface
 - Diet Recommendation Module
- Consolidated insights from over 20 research papers to support the system's design.
- Started initial layout testing and interaction flow verification.

8. Survey Paper Publication details

As part of the Major Project research workflow, our team initiated the development of a survey paper focusing on the integration of Ayurvedic dietary principles with modern cloud-based nutrient analysis systems. Although the paper has not yet been submitted for publication, significant groundwork has already been completed.

To establish a strong foundation, the team conducted an extensive review of literature from reputable journals, conference proceedings, and digital libraries. A total of 15 relevant research papers were shortlisted after screening numerous studies related to Ayurvedic diet systems, intelligent diet recommendation engines, cloud-based healthcare platforms, fuzzy logic models, and AI-driven nutrition analysis.

From these 15 papers, one research work was identified as the primary base paper due to its closest thematic alignment with our project objectives, methodological relevance, and potential for extension. This base paper currently forms the conceptual backbone for our proposed system's survey and problem framing.

The remaining components required for completing the survey paper—such as comparative analysis, structured taxonomy creation, methodological critique, gap identification matrix, and final synthesis of findings—are currently in progress. Formal writing, formatting, and validation for publication (IEEE/Elsevier/Scopus) will be completed as the project advances into subsequent phases.

Upon completion of the remaining survey components, the paper will be appropriately formatted according to the selected journal or conference guidelines and prepared for submission.

9. Conclusions:

In conclusion, AyurDietOS addresses a significant gap in modern Ayurvedic diet management by providing a cloud-based, integrated platform that combines traditional Ayurvedic principles with modern nutrient analysis. The project demonstrates how technology can streamline the process of creating personalized, dosha-aligned diet charts, manage patient profiles efficiently, and maintain a comprehensive database of foods and recipes with both nutritional and Ayurvedic properties. By automating diet chart generation and providing printable reports, the system reduces manual effort, minimizes errors, and enhances the overall accuracy and consistency of dietary guidance.

The development of AyurDietOS also highlights the feasibility of leveraging cloud technologies and modern web development tools in a student project setting, with minimal or no cost. It provides a foundation for future enhancements, including mobile integration, patient monitoring features, and potential integration with hospital management systems. Overall, the project demonstrates the practical application of technology in improving healthcare workflows, supporting Ayurvedic practitioners, and delivering holistic, scientifically informed dietary care to patients.

The successful implementation of AyurDietOS reinforces the importance of combining traditional knowledge with modern tools to create innovative solutions in health and wellness, ensuring that patients receive personalized, accurate, and holistic dietary guidance efficiently.

10. References:

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